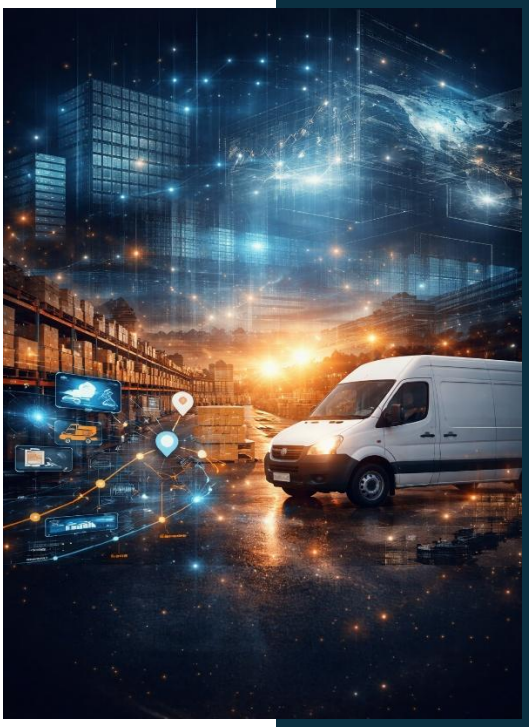
CMPG 311 Project

Attendance System



Group 7 [Phase 1]

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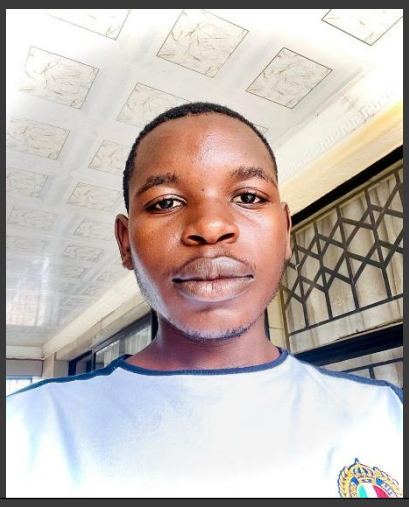

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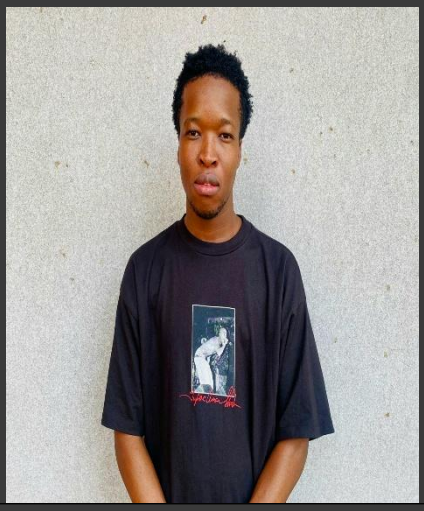

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1. Company Situation Analysis

1.1 Overview

Liftex is a medium-sized, technology-driven B2C delivery company with a mission to provide reliable, trackable, and efficient last-mile delivery services across South Africa. The company is a B2C delivery company with employee pay based on a hybrid-incentivised model and company-owned vehicles for better control over fuel expenditure data.

This project proposes a database system designed to improve parcel tracking, employee accountability, and payroll accuracy. The system addresses key operational challenges such as parcel theft, time fraud, and fuel inefficiencies by implementing audit trails, automated attendance tracking, and performance-based data monitoring.

1.2 Mission

To provide a reliable, trackable and efficient B2C delivery service to South Africa, ensuring a high standard of delivery tracking to customers and compensating employees fairly for their work based on performance metrics such as fleet efficiency, hours worked, and parcels delivered.

1.3 Operations

The Liftex system will provide:

- **Centralized Order Management:** Orders from clients are processed via the Liftex web app and stored in a centralised database at Liftex headquarters accessible by each warehouse.
- **Automated Shift Management:** Liftex employees check-in and clock-in daily at their nearest Liftex warehouse and are assigned a daily delivery sheet with deliveries that pass along a daily rotating route along some of South Africa's major cities. Deliveries to outlying areas near these routes are assigned provided they are deliverable to areas within a major city route with deliveries above a specific quota
- **Audit-Ready Logistics:** These daily deliveries are subject to a set upper limit (i.e. employee cannot have more than a certain amount of deliveries per day and deliveries to outlying areas take lower priority over deliveries on the major city route.
- **Automated Shift Management:** Upon the end of their shift, and partial or complete success of their delivery list, employees check-out at any Liftex warehouse.
- **Performance-Linked Payroll:** Employees are paid a fixed hourly rate + a **productivity** incentive bonus for each successfully completed delivery. Employees' fuel usage is measured against the company's fleet average and employee is

rewarded with an **efficiency** incentive bonus if it lies below said fleet average. This bonus is tiered, with the employees accessing certain bonuses below certain thresholds of fuel expenditure.

- **Fuel Management:** Fuel expenditure is measured via on-board telematics devices such as tachograph drive cards for odometer readings (distance travelled), fuel fleet cards for fuel **purchased** and a CAN-bus port for fuel **consumed**.

1.4 Organisational structure

Role	Purpose
Operations (Driver)	Responsible for the physical delivery of parcels.
Fleet Admin	Manages drivers' AARTO violations, check certificates of fitness and vehicle licensing to eNaTis, as well as managing professional driving permits (PrDP).
Finance Admin	Calculates hybrid payrolls and incentive bonuses.
Legal Admin	Maintains the company's legal compliance with South African legal framework (POPIA, NBCRFLI etc...).
Customer support	Consumer- facing role that responds directly to consumer queries, and texts, aiming for consumer satisfaction, and providing consumers with tracking data.
Customer Success	Proactively maintains a good relationship with client retail stores through targeted meetings and discussions to address how their needs can be met.
Sales	Responsible for the identification of and negotiation with high-volume retailers to form delivery contracts.

2. Problems & Constraints

2.1 Problems

- **Theft/misplacement and missing data:** The absence of a robust, auditable transit log makes it difficult to tell which employee held responsibility of a parcel when it was misplaced or stolen, leading to financial loss and customer dissatisfaction.
- **Time theft and 'buddy punching':** Manual clock-ins may result in employees clocking-in for non-present employees (buddy-punching), clocking-out for hours they did not work, or if the time at which an employee clocks in/out is not automated – allow for the logging of incorrect check in/out times (e.g. employee records 8:00

check in when real-time is 09:00, or clocking-in when they are running unrelated errands) i.e. ‘time theft”, resulting in inflated payroll costs and unfair compensation.

- **Inefficient Bonus Calculation:** The manual calculation of complex, tiered efficiency and productivity bonuses is prone to error and lacks transparency for employees.
- **Legal Compliance Gaps:** Difficulty in tracking employee-specific legal compliance (PrDP expiry, AARTO points, CoF status) against company policies.

2.2 Constraints

Technical Constraints

- **Development**: System should be developed using Oracle SQL developer.
- **System languages**: System backend will be written in NetCore whilst the front-end is written in html/css.
- **Data Security**: All personal and financial data must be encrypted.

Business Constraints

- **Security**: System data must be encrypted, comply with POPIA act, and track driver AARTO violations
- **Operational constraints**: Drivers are not allowed to exceed a certain number of deliveries per day.
- **Budget**: Development will utilize open-source tools where possible, with no budget for third-party GIS/GPS tracking software.

3. Database System Specification

3.1 Objectives

The primary objectives of the Liftex database system are to enhance operational accountability, ensure accurate workforce management, and establish a secure, auditable foundation for all logistical and payroll processes.

1. **Establish a Complete Chain of Custody for Parcels**: To eliminate ambiguity surrounding parcel accountability, the system will implement a comprehensive audit trail in the form of a transit log. This log will link every parcel to a specific employee at all times during the delivery process. By treating data completeness as a critical

system requirement, the database will enforce strict referential integrity, ensuring no delivery event is left unassigned or undefined.

2. **Eliminate Time Theft Through Automated Attendance Verification**: To ensure accurate and fair payroll processing, the system will automate the employee check-in and check-out process. This will be achieved by integrating location-based verification (e.g., IP filtering or geo-fencing) to restrict attendance logging to company premises, coupled with biometric or visual identity verification (e.g., selfie capture) to prevent unauthorized "buddy punching." This automation will generate immutable, precise time-stamped records.
3. **Create an Immutable Audit Trail for All Critical Actions**: To maintain data integrity and accountability, the system will automatically generate a secure, unalterable audit log for all modifications to sensitive data, such as attendance records and parcel assignments. This log will track who made a change, what was changed, and when it occurred, ensuring full transparency and facilitating post-incident investigations.

3.2 Information required from the database

The Liftex database will serve as the central repository for all operational, logistical, and financial data required to run the delivery service efficiently. It will store comprehensive information across five key domains: employee records, legal compliance, parcel logistics, fleet management, and payroll processing. This structured data collection will enable automated payroll calculations, real-time parcel tracking, proactive compliance monitoring, and robust audit capabilities. These data include:

1. **Employee Personal Data**

The system will store each employee's full name, contact details, and designated home warehouse. A unique Employee ID will serve as the primary key, linking all related records to the correct individual.

2. **Legal Compliance Data**

The database will track compliance information including status indicators for Certificate of Fitness and Professional Driving Permit (PrDP) validity, along with PrDP expiration dates to facilitate renewal reminders. The total number of AARTO violations incurred by each employee will also be maintained.

3. **Parcel and Logistics Data**

Each parcel will be identified by a unique Order ID, with records capturing order status (complete or incomplete), delivery start and end timestamps, and destination location. This data will link parcels to specific employees, creating an auditable transit log.

4. **Fuel and Fleet management**

Vehicle records will include a unique Vehicle ID, registration details with expiration dates, and validity status. Fuel consumption data will capture odometer readings, fuel litres purchased, and fuel litres consumed, enabling efficiency bonus calculations based on comparison with fleet averages.

5. **Finance and payroll calculation**

The database will store employee hourly rates and track total parcels delivered and total hours worked derived from clock-in and clock-out timestamps. Warehouse IDs will verify check-in locations. This data enables automated calculation of base pay, productivity bonuses, and efficiency bonuses.

3.3 Scope

The system will primarily focus on tracking employee attendance and performance metrics throughout the course of their daily shifts as well as document legal compliance data for each employee.

3.4 Boundaries

The Liftex Delivery Intelligence System defines clear boundaries to ensure secure, efficient, and controlled access to operational data. Delivery drivers are limited to interacting only with their assigned delivery tasks, allowing them to view delivery details, update parcel statuses, and record completed deliveries. This restriction ensures accountability while preventing unauthorized access to sensitive company data.

Administrative roles, including Fleet, Finance, and Legal administrators, have broader system access based on their responsibilities. They are authorized to manage employee records, monitor delivery performance, track fuel usage, and oversee payroll calculations. By restricting critical operations such as payroll adjustments, compliance tracking, and system modifications to authorized personnel, the system maintains data integrity and reduces the risk of manipulation or fraud.

Additionally, the system does not handle inventory management from retail partners, customer payment processing, or real-time GPS tracking of drivers. These exclusions define the operational scope of the system and ensure that the database remains focused on delivery tracking, employee performance, and compliance monitoring.

3.5 Entity Definitions

Entity	Description
Employee	Stores personal details (name, ID, contact, warehouse assignment), and links to legal compliance, payroll, and delivery records.
Vehicle	Contains details for each company-owned vehicle (registration, type, fuel consumption data). Links to employees via DELIVERY_SHIFT.
Delivery_shift	Represents a single driver's daily shift. Links an EMPLOYEE to a VEHICLE and stores check-in/out timestamps and total deliveries completed.
Parcel	Stores individual parcel details (order ID, status, timestamps, destination). Crucially, it links to the DELIVERY_SHIFT to establish chain of custody.
Fuel Transaction	Logs each fuel purchase (litres, cost, date) and links it to a VEHICLE. This is used to calculate efficiency bonuses.
Payroll Record	Stores the calculated payroll for an EMPLOYEE for a given period, breaking down base pay, productivity bonus, and efficiency bonus.

3.6 Relationship & Constraints

- **One-to-Many:**
  - One employee can have many delivery shift records.
  - One vehicle can be used for many delivery shift records.
- **Many-to-Many:**
  - An employee can operate multiple different vehicles over time, depending on daily availability and vehicle assignment. Conversely, a single vehicle can be operated by multiple different employees across different shifts.
  - An employee may be required to check in at and be based out of multiple warehouses, particularly when delivering across different geographic regions. Conversely, each warehouse may serve as a base for multiple employees.
- **Data Constraints:**
  - Unique constraint on employee email.
  - Foreign key constraints linking parcel to delivery shift and delivery shift to employee. Validation to ensure the number of parcels in a shift does not exceed the defined operational limit.

3.7 Security & Performance Considerations

- **Authentication & Authorization:** Implement role-based access control. Drivers see only their shifts; Finance Admin sees payroll; Fleet Admin sees vehicle data.
- **Performance Optimization**: Indexes will be created on frequently queried fields like EMPLOYEE.employee_id, PARCEL.order_id, and DELIVERY_SHIFT.shift_date.
- **Data Backup & Recovery**: The database will have a scheduled backup strategy to prevent data loss